

David Manuel

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of enterprise engineering experience designing full-stack web applications, microservices, and AI-enabled platforms. Proven track record of architecting cloud-native solutions with JavaScript/TypeScript, React, Node.js, and Python, as well as integrating production AI/LLM pipelines (RAG, OpenAI, Claude APIs) and agentic development tools (Claude Code, Cursor). Expert in cloud infrastructure (AWS, Azure, Terraform), automated CI/CD pipelines, design system component libraries, and enterprise observability. Strong cross-functional partner adept at translating complex organizational requirements into scalable software. Seeking a full-time, remote Senior Software Engineer role.

CORE COMPETENCIES

Languages & Frameworks: TypeScript, JavaScript (ES6+), React, Node.js, Next.js, Python (FastAPI, Django), HTML5, CSS3, Tailwind CSS, Material UI

AI & Knowledge Systems: LLM Integration (OpenAI API, Claude API), RAG Architecture, Vector Databases (Pinecone), Agentic Workflows (Claude Code, Cursor)

Cloud & Infrastructure: AWS (EKS, ECS, Lambda), Microsoft Azure, Terraform, Docker, Kubernetes, CI/CD (GitHub Actions, Azure DevOps)

Databases & Architecture: PostgreSQL, MongoDB, Redis, Elasticsearch, REST APIs, GraphQL, Microservices, System Architecture

Observability & Security: Datadog, Prometheus, Grafana, OAuth2, JWT, Enterprise SSO, RBAC, Audit Logging, Security Protocols

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Dec 2022 – Present | Rochester, NY

Netsmartz

- Architected and maintained scalable full-stack applications and microservices using TypeScript, React, Node.js, and Python, supporting multi-tenant enterprise workflows and Knowledge & Innovation initiatives.
- Spearheaded production AI/LLM integrations, developing Retrieval-Augmented Generation (RAG) search pipelines, document analysis tools, and agentic workflows utilizing OpenAI, Claude APIs, and Pinecone vector databases.
- Reduced median API latency from 850ms to 380ms (32% performance gain) through asynchronous execution, query optimization, and Redis caching.
- Built polished frontend user interfaces and component design libraries in React and TypeScript, establishing foundational design system patterns across internal products.
- Managed infrastructure-as-code and cloud deployments across AWS and Azure using Terraform, Docker, and Kubernetes, maintaining robust automated CI/CD pipelines in GitHub Actions.
- Established comprehensive observability, logging, and monitoring practices using Prometheus and Datadog to maximize production system reliability.

- Partnered directly with product leads, data scientists, and non-technical legal/business stakeholders to gather feedback, conduct live product demonstrations, and prototype new capabilities.

Senior Full Stack Software Engineer

Feb 2020 – Nov 2022 | Chicago, IL

Tempus AI

- Developed secure full-stack web features and REST/GraphQL APIs utilizing Python (FastAPI), Node.js, JavaScript/TypeScript, and React for complex analytics and data platforms.
- Improved p95 backend API responsiveness by 24% through database indexing, schema tuning, and backend query optimization across PostgreSQL and Redis.
- Configured enterprise security controls (OAuth2, JWT, RBAC, field-level encryption, and audit trails) to comply with strict regulatory and security frameworks.
- Designed modular React frontend component suites for enterprise reporting modules and administrative dashboards.
- Containerized multi-service architectures and automated deployments to cloud targets (AWS/GCP) using Docker, Kubernetes, and CI/CD pipelines.

Software Engineer

Dec 2015 – Jan 2020

Talmate

- Engineered full-stack modules and RESTful/GraphQL API services (Node.js, JavaScript, Python) for workflow automation and third-party SaaS integrations.
- Cut average workflow execution times from 12 minutes to under 6 minutes (45% improvement) via backend performance tuning and system caching.
- Built automated build, test, and release CI/CD pipelines with GitHub Actions and Azure DevOps.
- Collaborated closely with cross-functional team members to prototype features, conduct code reviews, and drive engineering best practices.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2011 – May 2015 | Houston, TX

University of Houston